

Heavy Mechanical Complex Taxila

(ISO 9001 Certified)



Summer Internship Report 2026

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Introduction to HMC

Heavy Mechanical Complex (HMC), rightly known as “Mother Engineering Industry” is serving the nation since more than four decades. HMC was principally entrusted the task of shifting the Pakistan’s consumer goods industry to the capital & producer goods industry. Heavy Mechanical Complex (HMC) was established in 1971 as a strategic industrial initiative by the Government of Pakistan, with financial and technical assistance from China. Its purpose was to support the country's growing need for heavy engineering and industrial infrastructure. In 1977, a subsidiary facility called Heavy Foundry and Forge (HFF) was set up, also with Chinese cooperation, to handle the specialized requirements of metal casting and forging. Later in 1991, both HMC and HFF were merged into a single integrated industrial entity under the name Heavy Mechanical Complex, forming one of the most significant symbols of Pak–China industrial collaboration.

1. Location:

HMC is located on Hattar Road, Taxila, in the Rawalpindi District, approximately 30 kilometers from Islamabad. Its strategic position near the federal capital allows easy access to major industrial zones and logistics routes.

2. Quality Standards and Certifications:

- ASME (American Society of Mechanical Engineers): HMC follows ASME codes for boiler and pressure vessel manufacturing.
- ISO 9001 Certification: HMC is certified under ISO 9001, which ensures quality management systems are maintained.
- BNBI Standards and PNRI Standards were also mentioned as important benchmarks for national and industrial compliance, particularly in nuclear and energy sectors.

3. HMC Section:

- Heavy mechanical complex I (HMC I)
- Heavy forge and foundry work (HMC II)
- Heavy mechanical complex III (HMC III)

4. Product Planning and Control (PPC):

On the second day of our internship, we attended an informative presentation about the Product Planning and Control (PPC) Department. We learned that PPC is responsible for planning and organizing production activities, scheduling work, monitoring progress, and ensuring that resources are available when needed. The department helps maintain smooth production, meet deadlines, control costs, and ensure the required quality of the final products.

4.1 Sales Order Numbering System

During the presentation, we learned about the Sales Order Numbering System used at HMC.

The sales order uses a 6-digit format: **XXYYZZ**.

- **XX** = Code for the specific product or item category
- **YY** = Fiscal year code
- **ZZ** = Serial number of the order

The fiscal year at HMC ends on **30 June**.

This numbering system makes it easier to identify and track sales orders.

Example

If an order number is: **122501**

It can be understood according to the HMC numbering structure:

- **12** → Product/category code
- **25** → Fiscal year code
- **01** → First order in that sequence

4.2 Inventory Control Numbering System

HMC also uses an **8-digit Inventory Control Number** to identify materials.

The number is divided into three parts:

XX – XX – XXXX

First 2 digits: Represent the main material type, such as steel, copper, etc.

Next 2 digits: Represent the sub-group of that material.

Last 4 digits: Represent the serial identification related to the material, such as its size or standard.

This system makes it easier to:

- Identify materials
- Find materials in inventory
- Maintain records
- Track material usage
- Avoid confusion between different materials

5. Main Types of Work at HMC

5.1 Mechanical Works

- Includes metal fabrication, machining, and assembly of mechanical components.

5.2 Foundry and Forge

- Involves casting and forging of raw materials to form industrial-grade components and blanks.

6. Mechanical Workshops

Within the mechanical domain, two key workshops are involved:

- Fabrication Shop: Handles cutting, bending, and assembling of metal structures using techniques like welding and bolting.
- Machine Shop: Involves turning, milling, drilling, and finishing processes to achieve precise mechanical components.

7. Machine Shop-1:

In Machine shop there are 4 bays, these are as follows:

- 1) Small Bay
- 2) Medium Bay
- 3) Large Bay
- 4) Assembly Bay

7.1 Small Bay:

In small bay all the machines produce small workpieces as product, this bay contains all machines small in size as compared to the Medium & Large Bay. This Bay contains Shaper Machines, Planner Machines, Lathe Machines, Milling Machines and Slotting Machines.

7.1.1 Shaper Machine:

Following are the parts of this Machine:

Base, Column, Ram, Tool Head, Clapper Box, Table, Saddle, Cross Rail, Stroke Adjusting Mechanism.

Following are the operations performed on this Machine:

1. **Horizontal Shaping** – A flat horizontal surface is generated by reciprocating a single-point tool across the stationary workpiece.
2. **Vertical Shaping** – A vertical or angled surface is machined by orienting the tool slide vertically on the machine.
3. **Angular Shaping** – The tool head is swiveled to a set angle to cut inclined or beveled surfaces.
4. **Slot/Groove Cutting** – A narrow slot or groove is cut into the workpiece using a shaped or narrow-edged tool.
5. **Keyway Cutting** – An internal or external keyway is machined to allow shaft-hub locking.

7.1.2 Planer Machine:

Following are the parts of this Machine:

Bed, Table, Housings/Columns, Cross Rail, Tool Heads, Saddle, Driving Mechanism.

Following are the operations performed on this Machine:

1. **Horizontal Plane Surfacing** – Large flat surfaces are machined by moving the table (with the workpiece) beneath a stationary or slowly-fed tool.
2. **Vertical Surfacing** – Vertical faces of large workpieces are machined using a vertically mounted tool head.
3. **Angular Surfacing** – Inclined surfaces are produced by adjusting the tool head to the required angle.
4. **Slotting on Large Workpieces** – Slots or grooves are cut into large components too big for a shaper table.

5. **Multi-tool Machining** – Several tool heads work simultaneously on different surfaces of the same large workpiece to reduce cycle time.

7.1.3 Lathe Machine:

Following are the parts of this Machine:

Bed, Headstock, Tailstock, Chuck, Carriage, Cross Slide, Compound Rest, Tool Post, Lead Screw, Feed Rod, Apron.

Following are the operations performed on this Machine:

1. **Turning** – Excess material is removed from the outer diameter of a rotating workpiece to produce a cylindrical surface.
2. **Facing** – The end face of the workpiece is machined flat and perpendicular to its axis.
3. **Drilling** – A hole is produced along the axis of the rotating workpiece using a drill held in the tailstock.
4. **Boring** – An existing hole is enlarged and made more precise using a single-point boring tool.
5. **Threading** – External or internal helical threads are cut by feeding the tool in sync with the workpiece rotation.
6. **Knurling** – A patterned (diamond/straight) texture is impressed onto the surface for grip, using a knurling tool.
7. **Parting/Cut-off** – The finished workpiece is separated from the remaining stock using a narrow parting tool.

7.1.4 Milling Machine:

Following are the parts of this Machine:

Base, Column, Knee, Saddle, Table, Spindle, Arbor, Overarm, Ram.

Following are the operations performed on this Machine:

1. **Slab Milling** – A flat surface is produced using a cylindrical cutter with its axis parallel to the workpiece surface.
2. **Face Milling** – A flat face is machined using a cutter mounted perpendicular to the surface, removing material with the cutter face and periphery.
3. **Slot Milling** – A slot of specific width is cut into the workpiece using a side or end mill.
4. **Gear/Form Milling** – Gear teeth or other special profiles are generated using a shaped form cutter.
5. **Drilling/Boring on Mill** – Holes are produced or enlarged using the milling machine's spindle in place of a dedicated drill press.
6. **Angular Milling** – Angled surfaces or chamfers are cut by tilting the cutter or workpiece to the required angle.

7.1.5 Slotting Machine:

Following are the parts of this Machine:

Base, Column, Ram, Tool Head, Saddle, Table, Rotary Table Mechanism, Stroke Adjusting Mechanism

Following are the operations performed on this Machine:

1. **Internal Keyway Cutting** – Keyways are cut inside bores using the vertically reciprocating tool.
2. **Internal Slot Cutting** – Internal grooves or slots are machined that are inaccessible to a shaper.
3. **External Profile Slotting** – External irregular profiles are cut using the vertical ram motion combined with rotary table indexing.

4. **Splines Cutting** – Internal splines are generated on hubs or gears requiring axial engagement features.
5. **Contour/Die Cavity Cutting** – Internal cavities or die profiles are shaped for tooling and die-making work.

7.1.6 BVT Machine

Following are the parts of this Machine:

Bed, Headstock, Spindle, Chuck, Tailstock, Carriage, Cross-slide, Tool Post, Lead Screw, Feed Rod, Apron, Hand Wheel, Gearbox

Following are the operations performed on this Machine:

1. **Turning:** Removes material from the outer surface to produce a cylindrical shape.
2. **Facing:** Removes material from the end face to make it flat and smooth.
3. **Boring:** Enlarges and finishes an existing hole accurately.

7.2 Medium Bay:

In Medium Bay all the machines produce medium workpieces as product, this bay contains all machines medium in size as compared to the Small & Large Bay. This Bay contains Shaper Machines, Planner Machines, Lathe Machines, Gear Hobbling Machine, Milling Machines and Slotting Machines. Just the difference was that the machine sizes were different.

7.3 Large Bay:

In Large Bay all the machines produce large workpieces as product, this bay contains all machines large in size as compared to the Medium & Small Bay. This Bay contains, Planner Machines, Lathe Machines, Milling Machines. Just the difference was that the machine sizes were different.

7.4 Assembly Bay:

The following parts were being assembled in the Assembly Bay:

1. **Mill Cheek:** The mill cheeks were fitted on both sides of the mill to support the rollers, shafts, and other components. They provide the main structural support to the mill.
2. **Rollers:** The rollers were mounted between the mill cheeks and connected with the shafts. They rotate to crush and extract juice from the sugarcane.
3. **Scrapers:** Scrapers were fitted near the rollers to remove accumulated material and keep the roller surfaces clean during operation.
4. **Pinion:** The pinion was fitted with the corresponding gear to transmit rotary motion from the drive system to the rollers.
5. **Sprocket:** The sprocket was there; it is basically used to transmit power between rotating components.
6. **Brass Bearing:** Brass bearings were fitted at the shaft-supporting points to provide smooth rotation and reduce friction and wear.
7. **Dumped Grate:** The dumped grate was assembled in the required position to support and guide material while allowing unwanted material or residue to pass through.
8. **Shafts:** The shafts were fitted through the rollers and supported by bearings to transmit torque and rotational motion.

9. **SA Fans:** SA (Secondary Air) fans were assembled with their drive and ducting connections to supply secondary air for combustion.
10. **ID Fans:** ID (Induced Draft) fans were assembled with the boiler's flue-gas system to draw combustion gases out of the boiler and maintain proper furnace draft.

7.5 Tools Section:

1. **Flat Tool:** Used for producing flat horizontal surfaces on a workpiece.
2. **Side Cutting Tool:** Used for machining vertical or side surfaces.
3. **Round-Nose Tool:** Used for finishing operations and achieving a smoother surface.
4. **Parting/Slotting Tool:** Used for cutting slots, grooves, and keyways.
5. **Roughing Tool:** Used for quickly removing a large amount of material during initial machining.
6. **Finishing Tool:** Used for producing a smooth and accurate final surface.
7. **Side Tool:** Used for machining vertical or angled surfaces of large workpieces.
8. **Slotting/Grooving Tool:** Used for making slots and grooves in workpieces.
9. **Turning Tool:** Used for reducing the diameter of a rotating workpiece.
10. **Facing Tool:** Used for producing a flat surface on the end of a workpiece.
11. **Boring Tool:** Used for enlarging and finishing existing internal holes.
12. **Threading Tool:** Used for cutting external or internal screw threads.
13. **Parting Tool:** Used for cutting off a finished component from the remaining material.
14. **Knurling Tool:** Used for producing a patterned surface to improve grip.
15. **End Mill:** Used for slotting, profiling, and machining various surfaces.
16. **Slab (Cylindrical) Mill:** Used for producing flat surfaces through peripheral cutting.
17. **Face Mill:** Used for efficiently machining large flat surfaces.
18. **Slot Drill:** Used for cutting slots and pockets and can plunge directly into the workpiece.
19. **Form Cutter:** Used for producing specific shapes and profiles such as gear teeth.
20. **Flat/Square-Nose Tool:** Used for cutting internal slots and keyways.
21. **Round-Nose Tool:** Used for producing smooth internal profiles and finishing cuts.
22. **Single-Point Turning Tool:** Used to remove material from the outer surface of a workpiece.
23. **Facing Tool:** Used to produce a flat and smooth end surface.
24. **Vernier Caliper:** Used to measure outside dimensions, inside dimensions, and depths accurately.
25. **Vernier Height Gauge:** Used to measure and mark accurate heights on a workpiece.

- 26. **Vernier Depth Gauge:** Used to measure the depth of holes, slots, and recesses.
- 27. **Try Square:** Used to check whether two surfaces are at a 90-degree angle.
- 28. **Steel Rule:** Used for general-purpose measurement of length and dimensions.
- 29. **Measuring Tape:** Used for measuring larger dimensions and distances.

7.6 Tool Sharpening Section:

Cutting tools become dull with continuous use, which increases cutting force, heat, and vibration and can reduce machining accuracy. Sharpening is done by grinding the worn cutting edge on an abrasive wheel to restore its proper shape and sharpness. Proper sharpening improves surface finish, dimensional accuracy, cutting performance, and extends the tool's service life.

7.7 Marking Section:

In Marking, during our visit they were making on the drum head, they were marking an ellipse on it and few holes for the nozzles. The tools they were using are divider, height gage, vernier caliper, Tummel, steel ruler, try square, chowks, and red paint.

8. Lean Manufacturing:

Lean Manufacturing is a method of production that focuses on reducing waste and improving efficiency. It aims to use time, materials, machines, space, and manpower effectively while maintaining good product quality. The main goal of lean manufacturing is to remove unnecessary activities, reduce production costs, improve workflow, and deliver products to customers efficiently.

8.1 5S of Lean Manufacturing:

1. **Sort:** Remove unnecessary tools, materials, and items from the workplace and keep only what is required.
2. **Set in Order:** Arrange tools, equipment, and materials properly so they can be easily found and used.
3. **Shine (Seiso):** Keep the workplace, machines, and equipment clean to maintain a safe and efficient environment.
4. **Standardize:** Establish standard procedures for maintaining cleanliness, organization, and proper working practices.
5. **Sustain:** Develop discipline to follow the 5S practices regularly and continuously.

8.2 8 Types of Waste in Lean Manufacturing:

1. **Defects:** Products or components that contain errors and require repair, rework, or rejection.
2. **Overproduction:** Producing more products than required or producing them earlier than needed.
3. **Waiting:** Time wasted when workers, machines, or materials are waiting for the next operation.
4. **Non-utilized Talent:** Not properly using the skills, knowledge, ideas, and experience of employees.

5. **Transportation:** Unnecessary movement of materials, products, or equipment from one place to another.
6. **Inventory:** Keeping excessive raw materials, work-in-progress, or finished products.
7. **Motion:** Unnecessary movement of workers or equipment during the production process.
8. **Extra-processing:** Performing unnecessary operations or work that does not add value to the product.

9. Fabrication Shop

Fabrication refers to the process of converting raw metal stock plates, sheets, and structural sections into finished components or assemblies through cutting, forming, welding, and joining operations. Its purpose in mechanical engineering is to build structural and functional parts that meet dimensional, strength, and design requirements. It bridges raw material procurement and final assembly by shaping metal into usable engineering forms. Fabrication is essential wherever custom or heavy components cannot be economically produced through casting or machining alone.

Fabrication Shop is divided into several sections; these sections are as follows”

- 1) Welding Section
- 2) Layout Section
- 3) Hot Forming Section
- 4) Press Section
- 5) Rolling Section

9.1 Welding Section:

Submerged Arc Welding (SAW): Submerged Arc Welding is a welding process in which an electric arc is formed between a continuously fed wire electrode and the workpiece. The arc and molten weld pool are covered by a layer of granular flux, which protects the weld from atmospheric contamination. It is mainly used for **long, straight welds and heavy steel components** because it provides high welding speed, good penetration, and a smooth weld finish.

9.2 Layout Section:

1. **Plasma Cutting:** Plasma cutting uses a high-velocity ionized gas jet to melt and expel metal along a marked path, producing a clean, narrow cut suitable for various plate thicknesses. It is commonly used for shaping profiles on conductive metals prior to further processing.
2. **CNC Gas Cutting:** CNC gas cutting employs computer-controlled torches using oxy-fuel flame cutting to achieve precise, repeatable profiles on thicker steel plates. Programmed cutting paths improve dimensional accuracy and reduce manual marking errors.
3. **Auto-Semi Plasma Cutting:** This process combines automated and manually guided plasma cutting to handle a range of plate sizes and profile complexities. It offers a balance between cutting speed, precision, and operator control for varied fabrication requirements.

9.3 Hot Forming Section:

1. **TIG Welding:** Tungsten Inert Gas (TIG) welding uses a non-consumable tungsten electrode with an inert shielding gas to produce high-quality, precise welds. It is typically used where weld appearance, control, and joint accuracy are critical.

2. **MIG Welding:** Metal Inert Gas (MIG) welding uses a continuously fed consumable wire electrode with shielding gas to join metal parts efficiently. It is favored for its higher deposition rate and suitability for medium-to-thick sections.
3. **Electric Welding:** Electric (arc) welding generates heat through an electric arc between an electrode and the workpiece to fuse metal at the joint. It is a widely used, versatile process for structural and general fabrication welding.

9.4 Press Section:

Press Machines for Plates and Sheets: Press machines apply controlled mechanical or hydraulic force through dies to bend, form, or shape metal plates and sheets into required geometries. This process enables the production of structural and shaped components with consistent dimensional accuracy.

9.5 Rolling Section:

Rollers for Sheets and Plates: Rolling machines pass metal plates or sheets between rotating rolls to progressively bend or reduce thickness, producing curved or flat formed sections. This process is essential for shaping large plates into cylindrical or contoured components used in heavy fabrication work.

10. Forging Shop

During our internship at Heavy Mechanical Complex (HMC), Taxila, we visited the Forge Shop to understand the basic principles and equipment used in metal forging. Forging involves heating metal to a suitable temperature and then shaping it by applying high mechanical forces. The shop contains heavy-duty equipment such as steam hammers, hydraulic presses, forging dies, and trimming presses, which are used to manufacture strong and durable components for various engineering applications.

10.1 Steam Hammer Forging:

The steam hammer was one of the major machines observed in the Forge Shop. It works by using steam pressure to raise a heavy ram, which is then allowed to fall onto the heated workpiece. The repeated impact forces gradually deform the hot metal and produce the required shape.

The Forge Shop has four steam hammers with different capacities:

1. One 1-ton steam hammer
2. Two 2-ton steam hammers
3. One 3-ton steam hammer

These machines are selected according to the size and type of component being forged.

10.2 Boiler System

A continuous supply of steam is required to operate the steam hammers. For this purpose, the Forge Shop has a boiler system that generates the required steam.

The boiler observed during the visit was an **oil-gas-fired packaged steam boiler** with an evaporating capacity of **10 tons per hour**. Its serial number was **23106**, and it was commissioned in **January 1977**. The Forge Shop has four boilers in total; at the time of our visit, three were operating while the one of them was under maintenance.

10.3 Water Softening Plant

The boiler system is also supported by a water softening plant. Natural water contains minerals such as calcium and magnesium that make it hard. When hard water is supplied to a boiler, these minerals can form deposits on the boiler tubes and reduce heat-transfer efficiency.

The water softening process helps to reduce hardness and provides better-quality water for boiler operation. It helps minimize scaling, reduce maintenance requirements, protect boiler components, and improve the overall service life of the boiler.

10.4 Pre-Forging and Die Section

Before the final forging stage, the heated metal may undergo pre-forming to bring it closer to the required shape. Forging dies are then used to control the final shape and dimensions of the component.

The general sequence of the process is:

Heating → pre-forming → Die Forging → Final Shape

Pre-forging makes the final operation easier and can also help reduce material wastage.

10.5 Trimming Press

After the forging operation, some excess material known as **flash** may remain around the edges of the forged component. A trimming press is used to remove this unwanted material.

During our visit, we observed a trimming press having a maximum capacity of approximately **1000 tons**. It helps remove excess flash, improve the appearance of the forged component, and prepare it for subsequent machining processes.

10.6 Hydraulic Press

Hydraulic presses are also used in the Forge Shop for heavy forming operations. Unlike a hammer, a hydraulic press applies a controlled and continuous force to the heated workpiece.

During our visit, we observed the **3007 hydraulic press**, which is used for specialized forming operations. Hydraulic pressing is particularly useful when a gradual and accurately controlled force is required.

The Forge Shop plays an important role in heavy engineering by producing components with improved strength, durability, and reliability. Different forging techniques and machines are used depending on the required shape and size of the component.

The major activities carried out in the shop include **heating, pre-forming, hammer forging, die forging, hydraulic pressing, and trimming**. Together, these operations allow a wide range of heavy-duty forged components to be manufactured for industrial applications.

11. Boiler House:

An industrial boiler is a closed pressure vessel used to heat water and convert it into steam by transferring heat from a fuel source. The generated steam is supplied at a controlled pressure and

temperature for different industrial applications. In the Forge Shop, the steam produced by the boilers is mainly used to operate the steam hammers required for forging operations.

During our internship, we observed that the Boiler House at HMC contains four boilers, out of which three were operational and one was under maintenance. Each boiler has a capacity of approximately 10 tons and produces steam at an output pressure of about 6 bar and a temperature of approximately 250°C. The boilers provide a continuous steam supply to the steam hammers in the Forge Shop.

The basic working process of the boiler starts with feeding water into the boiler, where it is heated using the heat produced by fuel combustion. As the water absorbs heat, it changes into steam, which is collected and supplied through steam outlet pipes. The steam is delivered to the steam hammers at the required pressure and temperature. Safety equipment, including pressure control and relief systems, is used to maintain safe boiler operation.

12. Machine Shop-2:

In Machine shop there are 4 bays, these are as follows:

- 1) Small Bay
- 2) Medium Bay
- 3) Large Bay

All of the machines, present in Machine Shop has been discussed in Machine Shop 1. The difference was that the Machine shop 2 mainly do maintenance of machines or do large projects. We have seen there were big shafts as compared to Machines shop 1 and the diameter of rollers were also large as compared to the Machine shop 1.

13. Pattern Shop:

The Pattern Shop is responsible for producing patterns used in sand-mold preparation for the casting process. Patterns replicate the external geometry of the final component, with additional allowances incorporated to compensate for shrinkage, machining, and other dimensional effects encountered during casting and cooling.

13.1 Planer Machine:

In pattern making, a planer machine is used to true and smooth the flat surfaces of wooden pattern stock before shaping. It removes excess material to achieve flat, parallel, and dimensionally consistent surfaces, providing an accurate base from which the pattern is further shaped and finished. This ensures the resulting mold cavity maintains proper dimensional integrity.

13.2 Machining Allowance:

Machining allowance is extra material added to the pattern dimensions to account for the metal that will later be removed during finishing operations such as turning, milling, or grinding. It ensures the cast component retains adequate stock for achieving the required dimensional accuracy and surface finish after machining.

13.3 Shrinkage Allowance:

Shrinkage allowance is provided by making the pattern slightly larger than the finished component to compensate for the contraction of metal as it cools from solidification temperature to room temperature. Since different metals shrink at different rates, this allowance is essential to ensure the final casting matches the intended dimensions.

13.4 Taper Allowance:

Taper allowance (draft) is a slight inclination given to the vertical faces of a pattern to allow it to be withdrawn from the sand mold without damaging the mold cavity. Without adequate taper, removing the pattern could cause mold breakage or dimensional distortion in the cavity walls.

13.5 Print:

A print refers to a projection added to the pattern that creates a corresponding seat or impression in the mold, allowing a core to be accurately positioned and supported during the casting process. It ensures correct core alignment and prevents shifting during metal pouring.

13.6 Riser:

A riser is a reservoir of molten metal connected to the casting cavity, designed to feed additional liquid metal into the casting as it solidifies and contracts. Since metal shrinks during cooling, the riser compensates for this volumetric reduction, preventing internal voids, shrinkage cavities, and porosity in the finished casting.

13.7 Core Burst:

A core burst occurs when the gases generated inside a core during pouring cannot escape properly and build up excessive pressure. This pressure buildup can cause the core to crack or break apart, releasing gas suddenly into the molten metal. As a result, the casting may develop internal voids, blowholes, or surface defects, compromising its structural integrity and dimensional accuracy.

13.8 Cope:

The cope is the upper part of the molding box (flask) in a sand mold. It houses the upper half of the mold cavity and typically contains the sprue and riser passages through which molten metal is poured and gases escape.

13.9 Drag:

The drag is the lower part of the molding box, resting on the molding floor or base. It contains the lower half of the mold cavity and provides the foundation on which the mold assembly is built.

13.10 Cheek:

The cheek is an intermediate section placed between the cope and drag, used only when the pattern geometry is too complex to be accommodated in a simple two-part (cope and drag) mold. It allows the mold to be divided into three parts, making it possible to withdraw patterns with intricate or multiple parting lines.

13.11 How They Fit Together:

The drag is prepared first and placed at the base, followed by the cheek (if required) positioned above it, and finally the cope is assembled on top. Together, these sections are aligned using locating pins and clamped or weighted to prevent shifting during pouring. Once assembled, the internal cavity formed between these sections defines the exact shape of the casting, with molten metal poured through the sprue in the cope to fill the cavity.

14. Cast Iron Shop:

Casting is a manufacturing process in which molten metal is poured into a mold cavity of the desired shape and allowed to solidify to form a component. Its basic purpose is to produce components with complex geometries that would be difficult or uneconomical to achieve through other manufacturing methods such as machining or forging.

14.1 Silica Sand:

Silica sand is widely used for mold preparation because of its high refractoriness, allowing it to withstand the high temperatures of molten metal without breaking down or fusing. It also has good permeability, allowing gases generated during pouring to escape, and sufficient bonding strength (with a suitable binder) to retain the mold shape during handling and pouring.

14.2 Facing Sand:

Facing sand is a fine-grained, high-quality sand applied directly around the pattern surface and mold cavity, in contact with the molten metal. It improves the surface finish of the casting and reduces surface defects, since this layer directly determines the texture and quality of the casting surface upon solidification.

14.3 Induction Furnace:

An induction furnace is used to melt cast-iron scrap by generating heat through electromagnetic induction. An alternating current passed through a coil surrounding the furnace induces eddy currents within the metal charge, and the resulting resistive heating melts the material. This method offers good control over melting temperature and metal composition.

14.4 Cupola Furnace:

A cupola furnace is a traditional shaft-type furnace used for melting cast iron on a continuous basis. It operates by charging alternating layers of coke, limestone, and metal (pig iron/scrap) from the top, while air is blown through tuyeres near the bottom to combust the coke and generate the heat required for melting. It is commonly used in foundries for large-scale, continuous cast-iron melting operations.

14.5 Electric Furnace:

An electric furnace melts cast-iron scrap using heat generated through electric arcs or resistance elements, depending on the furnace type. This method allows for precise temperature control and composition adjustment, making it suitable for producing cast iron with consistent metallurgical properties.

14.7 Cast Iron Temperature:

Cast iron generally melts within a temperature range of approximately 1150°C to 1200°C, while typical pouring temperatures for foundry practice range from about 1300°C to 1400°C, depending on the casting requirements. The exact pouring temperature depends on factors such as the specific cast-iron grade, casting size, mold design, and individual foundry practice, and should not be treated as a fixed industry-wide value.

15. Steel Foundry Shop:

15.1 Electric Arc Furnace:

The Electric Arc Furnace is used in the steel foundry shop for melting steel scrap. It has a maximum capacity of approximately 15 tons. The furnace operates through a PLC-based control system for monitoring and controlling the process. Electric arcs generate the required heat to melt the scrap. Argon gas is supplied during the process to help remove dust and impurities from the scrap. After melting, the molten metal is prepared for further casting operations.

15.2 Green Sand:

Green sand is commonly used for light casting applications. It is a mixture of sand, clay, water, and other additives that provide the required molding properties. It is easy to prepare, shape, and reuse for suitable casting operations.

15.3 Silica Sand:

Silica sand is mainly used for heavy casting applications where higher strength and temperature resistance are required. It can withstand the high temperatures of molten metal during casting. It provides good mold stability and helps produce accurate casting shapes.

15.4 Induction Furnace:

The Induction Furnace is used to melt metal through electromagnetic induction. It operates using a magnetic field generated around the furnace, with the electromagnetic effect producing heat in the metal. The furnace has a maximum capacity of approximately 1000 kg. It provides controlled and efficient melting of metal for foundry applications.